

Do CFLs Save Whole-House Energy?

If CFLs decrease cooling loads compared to incandescents, but increase heating loads, do they save energy overall?

by Anil Parekh

Energy-efficient compact fluorescent lamps (CFLs) are becoming more and more widely available in the marketplace. Using these energy-efficient lamps in homes reduces electrical energy use and power demand. However, it also affects space-heating and space-cooling energy use. The take-back effect—the interaction of lighting with other loads—alters the overall cost savings associated with lighting. During the heating season, CFL lighting reduces internal gains compared to incandescents and therefore increases the heating load. During the cooling season, CFL lighting reduces the cooling load. Do these effects offset each other? Is the net effect of using CFLs in place of incandescents positive or negative?

A number of field studies of commercial buildings show that replacing conventional lighting with energy-efficient lighting increases the heating load and reduces the cooling load. But residential use of lighting is a different story. In residential buildings, lighting is used selectively, depending on the occupants' needs. To learn more about the net impact of CFL lighting in housing, Natural Resources Canada performed detailed field research on the effect of CFL lighting on a home's overall energy use. We also sought to validate an internal-gains model associated with lighting energy use.

Average Lighting Energy Use in Canadian Homes

Over the years, several detailed housing surveys have been conducted in order to learn about energy use in Canadian homes. From a set of 134 highly monitored houses, profiles were generated to define lighting energy use in housing (see Table 1). The data from the monitored houses showed that lighting energy use accounts for 5%–8% of annual utility costs in cold-climate regions.

Twin House Test Facility

The Canadian Centre for Housing Technology (CCHT) maintains a facility consisting of two research houses, which provide a unique opportunity for examining the differences between different house technologies. Each is a typical two-story wood-framed house, with 2,260 square feet (210 square meters) of livable area, set on a cast-in-place concrete basement, with style and finish representative of new houses currently available in the local housing market. The houses are built to meet the R-2000 Standard (Canada's Energy Star equivalent), with a construction package that includes tight, well-insulated assemblies, and low-e argon-filled

sealed glazing units. Each house has a high-efficiency sealed-combustion condensing gas furnace, a power-vented conventional gas water heater, a gas fireplace, and a heat recovery ventilator (HRV). The furnace, water heater, and gas fireplace are all vented through the wall, eliminating the need for chimneys.

Each house also has the major appliances typically found in North American homes. In the research houses, human activity is simulated by a system that operates appliances, lighting, and other equipment according to a set schedule. This schedule is identical in both houses. The simulated occupancy system is also used to monitor energy performance.

Baseline Tests with Incandescent Lighting

The two research houses were set up and operated under identical conditions to develop a full profile for a reference. The lighting power demand for the two houses followed a nearly identical pattern, with a less than 0.4% difference. Although the normal load—that is, the lighting load with incandescents—was 3.4 kWh per day, we increased it to 10.2 kWh per day. The target for our research was to reduce the load by 7 kWh per day with CFL lighting. The reason we increased the lighting load to 10.2 kWh per day was to create a difference between the load with incandescents and the load with CFLs that was sufficiently great to be measured accurately. Our sensitivity analysis for the whole-house energy monitoring showed that our measurements were within $\pm 1\%$ when the difference between the two loads was greater than 5 kWh per day.

HEATING SEASON. We conducted two sets of baseline tests in March 24 and 25, 2004, and December 24 to 28, 2004 using incandescent lighting in both houses. The measured data showed that the difference between the two



loads was within 0.05 kWh per day, or about 0.5%. The space-heating energy use, measured in natural gas consumption, showed that the two houses were operating close to identically, with a difference of approximately 1.4 kWh, or 1.2%, per day.

COOLING SEASON. For a period of four days during the month of July 2004 we conducted tests to determine baseline cooling loads, using incandescent lighting in both houses. Cooling energy use was approximately 2.2 kWh per day during the test period. The difference in energy use between the two houses was less than 4%.

Testing with CFLs

We designated one of the two research houses as the test house and the other as the reference house. We kept the incandescents in the reference house, but we retrofitted the test house with CFLs. Thirty-one incandescent lamps were replaced with CFLs having a similar lumen output. Thus, for example, a 40W incandescent was replaced with a 9W CFL. We then used the simulated occupancy system to control lights and HVAC equipment.

HEATING SEASON. The 14-day test period, (March 26–28, April 15–18, and December 30 and 31, 2004; and January 1–5, 2005) covered the full range of heating loads, making it possible to compare the effects of the CFL lighting on heating load. We tested the houses with continuous ventilation and intermittent ventilation to see if that effected energy use. Space-heating energy used varied depending on outdoor conditions. It ranged from approximately 23 kWh to 120 kWh per day, representing 15%–100% of the space-heating load for the house. Our energy use comparison of heating load with incandescent and CFL lighting revealed the following:

The CFL lighting in the test house reduced electricity consumption by approximately 7.3 kWh per day and electricity used for lighting by about 68% per day.



Space-heating energy use in the home with CFLs increased to compensate for the reduction in lighting energy use. This increase represented 8%–33% of the daily space-heating load.

It appeared that the different ventilation strategies, either continuous or intermittent, did not have any impact on the overall energy savings associated with CFLs.

The reduction in lighting energy use in the home where incandescents were changed to CFLs was almost all offset by the increase in space-heating energy use.

COOLING SEASON. For the cooling season, we did the testing using calibrated CFLs and incandescent lamps and the same lighting schedule. Tests were conducted from August 10 to August 30, 2004, for a period of 21 days. During this time, daily maximum outdoor temperatures ranged from 68°F to 85°F (20°C–29.3°C). We observed the following trends:

The energy load for the air conditioning system was reduced by 2.1–3.8 kWh per day in the test house, as compared to the reference house. The amount of reduction depended on ambient conditions. On average, the difference was approximately 3.1 kWh per day for the test period.

The tests showed that approximately 78% of the internal gains due to lighting added to the cooling load.

The air conditioning system operated 330–830 minutes (5.5–13.8 hours) per day in the reference house and 200–752 minutes (3.3–12.5 hours) per day in the test house. Use of CFLs reduced the on-time run of the cooling equipment by 20% or more.

The tests showed that CFLs reduced the lighting load by 7.3 kWh per day and reduced the air conditioning load by 3.1 kWh per day. This represents a

total reduction in electrical energy use of 10.4 kWh per day for the test period.

Potential Energy Savings with CFLs

What is the overall impact of using CFLs instead of incandescents in homes located in different climates? Using HOT2000 Residential Energy Simulation program, we developed a range of house types and evaluated them for 33 different climates—11 in Canada and 22 in the United States. Our analysis was based on the following assumptions:

We assumed a typical new two-story house with about 2,000 square feet (186 m²) of heated floor area. The thermal archetype was based on age, location, and type of house (size, insulation levels, and airtightness; and heating, hot water, ventilation, and cooling systems). We assumed that during the heating season, each house is maintained at 70°F (21°C) on the main floors and 66°F (19°C) in the storage and basement rooms. During the cooling season, the houses are maintained at 77°F (25°C) throughout.

Energy consumption for incandescent lighting was assumed to be 3.4 kWh per day, although in fact energy consumption for lighting varies throughout the year, depending on the length of the days and on outdoor and indoor daylight levels.



Table 1. Lighting Energy Use in Single-Family Homes

Number of Occupants	Average Lighting Energy Use, kWh/day	Standard Deviation, kWh/day
1	1.6	0.4
2	2.6	0.7
3	3.0	0.8
4	3.2	0.6
5	3.4	0.6
6	3.5	0.7
7	3.7	0.3

TABLE 2. ENERGY AND COST SAVINGS ASSOCIATED WITH CFLS*

	**Space Heating Increase—kWh	Space Cooling Decrease—kWh/yr.	Cost Savings—Lighting Only—\$/yr	Overall Cost Benefits—\$/yr.	Take-back Effect
Vancouver	230	20	20	12	60%
Edmonton	293	38	26	22	85%
Saskatoon	262	59	29	24	83%
Winnipeg	272	61	19	11	58%
Toronto	241	63	27	22	81%
Sudbury	222	26	27	10	37%
Ottawa	251	53	27	21	78%
Montréal	182	54	20	12	60%
Quebec City	184	55	20	12	60%
Saint John	233	60	24	11	42%
Halifax	270	52	31	20	65%
St. John's	270	40	28	8	25%
Fairbanks, AK	304	18	38	27	71%
Los Angeles	52	72	38	44	116%
San Francisco	147	34	38	36	95%
Denver	188	37	26	20	77%
Miami	209	96	27	35	130%
Chicago	188	38	27	21	78%
Boston	209	32	37	32	86%
Minneapolis	209	33	25	18	72%
New York	178	40	45	43	96%
Madison, WI	209	34	28	21	75%
Seattle	220	25	20	12	60%

*Based on 318 kWh lighting energy reductions per year

** Natural gas and fuel oil converted to equivalent kWh

We assumed replacement of five incandescent fixtures with CFL fixtures. Based on this scenario, five fixtures with 77W incandescent lamps used for three hours per day are replaced with 19W CFLs. The reduction in lighting energy use is approximately 0.9 kWh per day.

We based our cost estimates on annual average residential unit energy costs for electricity and natural gas for the year 2005 for various locations.

Based on these assumptions, we generated detailed house models using HOT2000 energy analysis software (see Table 2). We observed the following trends:

The electrical energy savings are about 318 kWh per year with CFL

lighting. The reduction in lighting energy consumption is approximately 26%. The electrical demand savings are approximately 0.3 kW.

For heating-dominated regions, the increase in space-heating energy consumption is 0.6%–1.7% with CFL lighting.

For cooling-dominated regions, the reduction in space-cooling energy consumption is 4%–9.5% with CFL lighting. The reduction in on-time operation of cooling equipment is 15%–22%.

The take-back effect—that is, the interaction of lighting with other loads—significantly reduces estimated overall energy cost savings. For example, for a new house located in

Vancouver without air conditioning, estimated energy cost savings are approximately \$9 with CFLs compared to incandescent lighting. This includes cost savings associated with lighting and additional costs associated with space heating. The cost savings associated with lighting are \$20, but the additional cost of space heating is \$11. For an air conditioned house in Miami, the estimated energy cost savings are approximately \$35. Of this amount, \$8 is attributable to reduced air conditioning loads with CFLs compared to incandescents. If only lighting energy cost savings were considered, the overall savings would be approximately \$27.

Do CFLs Save Energy?

Our analysis shows the importance of a realistic evaluation of CFL lighting benefits in Canadian and U.S. homes. Most lighting suppliers and manufacturers advertise the energy savings associated with CFLs. But the figures that they quote are based only on the energy savings associated with the lamp itself. In heating-dominated climates, there is a significant take-back effect associated with CFLs. The analysis showed that the take-back effect could reduce cost savings by up to 40%. In cooling-dominated climates, on the other hand, the additive effect—that is, reduced air conditioning load—can increase cost savings by up to 30%.

But our conclusion is that CFLs do save energy overall. **H_e**

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For more information:

Visit the Web site of the Canadian Centre for Housing Technology (CCHT), www.ccht-cctr.gc.ca, for a full description of the twin-house test facility and a full report of the study described in this article.

Download the HOT2000 Residential Energy Simulation program at no cost from www.sbc.nrcan.gc.ca.

For more information on R-2000 Standard, go to <http://oee.nrcan.gc.ca/r-2000>.